Jaavin Mohanakumar

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Canadian Citizen (TN Eligible)

Education

McMaster University – BASc, Computer Engineering April 2027
Relevant Coursework: Digital Systems Design, Data Structures and Algorithms, Signals and Systems, Electronic Devices

Experience

Software Engineering Intern – IPEX Management Inc. – Oakville, ON May 2024 – Aug 2024
- Designed and implemented monitoring system in C# with alerts for machine failures, cutting response time by 30%
- Architected network configuration server with automated DHCP management supporting 200+ corporate users

Software Engineering Intern – IPEX Management Inc. – Oakville, ON May 2023 – Aug 2023
- Implemented file transfer service in C# using .NET for business-critical process, reducing downtime from 14% to 2%
- Improved API Gateway item-vendor mapping accuracy by 183 basis points and planned MongoDB migration

Selected Projects

Hardware Image Decompressor – (Technical Report) Oct 2025 – Dec 2025
- Designed and implemented RTL image decompressor achieving 88.89% multiplier utilization at 50 MHz
- Developed end-to-end decoder with IDCT, 10-tap FIR upsampling, and YUV-to-RGB conversion for 192×144 image

Real-Time 3D LiDAR Mapping System – (Technical Report) Feb 2025 – April 2025
- Created LiDAR system with MSP432 microcontroller, ToF sensor, and stepper motor achieving 360° spatial scanning
- Developed Python control panel with Open3D visualization for real-time 3D point cloud rendering

What's in Front of Me (WIFOM?) – (Youtube Demo) Jan 2023 – April 2023
- Designed wearable ML-powered object detection system with dual ESP32-CAM providing real-time audio and haptics
- Developed end-to-end pipeline integrating image capture, cloud-based ML processing, and haptics for under $40 CAD

MeGPT – (Website) Sept 2023 – Dec 2023
- Launched AI SaaS platform reaching 2.2K visitors in 24 hours; featured on McMaster news and startup accelerator

Student Design Teams

Controls Project Manager – McMaster Rocketry Team – Hamilton, ON Nov 2022 – Dec 2023
- Led controls sub-team of 15 members to design avionics and control systems for Spaceport America in New Mexico
- Coordinated across 6 sub-teams and 70+ members to integrate flight control, telemetry, and data acquisition

Software Development Co-Lead – FIRST Robotics Team 1325 – Mississauga, ON Sept 2021 – June 2022
- Designed autonomous control systems in Java implementing path following, encoder/gyro feedback, and computer vision based targeting with trigonometric interpolation for 120lb competition robot
- Developed state machine control architecture with PID loops, feedforward gravity compensation, and sensor fusion enabling Excellence in Engineering Award at FIRST Robotics World Championship in Houston

Skills

- **Hardware:** RTL Design & Verification (Verilog/SystemVerilog), FPGA/ASIC Development, Digital Signal Process, Timing
- **Tools:** Quartus, C, C++, Java, Modelsim, ARMv7, Python, Git, Oscilloscope, OrCAD, MATLAB, Linux, Logic Analyzer
- **Protocols:** I2C, SPI, CAN, UART, USB (2.0), VGA, TCP/IP, UDP, DHCP, AXI, FIFO, PCIe

Portfolio

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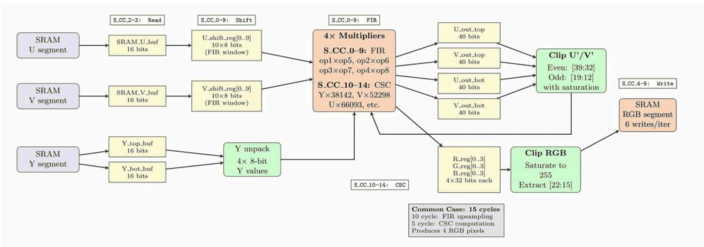
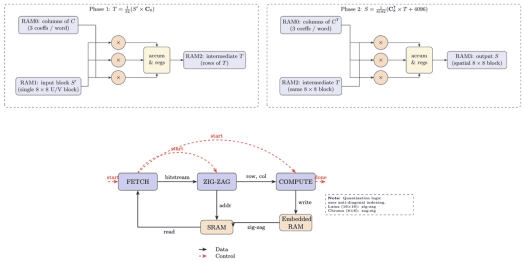
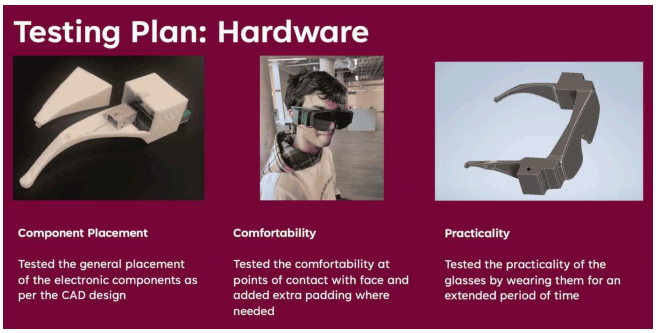
Hardware Implementation of an Image Decompressor, based on JPEG





- Architected RTL image decompressor for JPEG-like format achieving 88.89% hardware constrained multiplier utilization at 50 MHz, implementing 10-tap FIR upsampling and YUV-to-RGB colour space conversion with minimal stalls across the 192×144 image.
- Designed 4-stage pipelined IDCT processor (FETCH → COMPUTE_T → COMPUTE_S → WRITE) with time-shared hardware multipliers across 325 blocks; implemented column grouping to maximize parallel computation throughput.
- Optimized datapath architecture by deriving row/column indices combinationally from address counters and streaming final S-phase samples, eliminating redundant embedded RAM usage and reducing register overhead by 30%.
- Implemented circular buffer architecture with dual-port embedded RAMs for concurrent multiply-accumulate operations; achieving timing closure with 3.282 ns slack while utilizing only 4% of target FPGA logic elements.
- Validated RTL against Python reference model using unit-isolated testbenches with golden SRAM comparisons and no-write-region assertions covering boundary conditions (row-edge chroma filter, clipping). Debugged VGA through mismatch-address and ModelSim waveform tracing.
- **Technical documentation available here**

What's in Front of Me (WIFOM?) / Assistive Device for Visually Impaired



- Architected end-to-end computer vision pipeline for assistive wearable device with dual ESP32-CAM modules, web application middleware, and cloud-based ML inference backend, enabling real-time object detection and hazard identification for visually impaired users for under $40 CAD.
- Developed a prioritized alert system with text-to-speech auditory feedback and haptic motor actuation based on threat classification (dangers, warnings, objects), enabling multimodal sensory substitution with <2s latency from image capture to user notification.
- Implemented RESTful API integration layer connecting embedded ESP32 image capture to cloud ML service.
- Validated system robustness through field testing in high-density environments (crowds), occlusion scenarios (direct sunlight) demonstrating reliable multi-object detection.
- **Technical slides available here and video demonstration available here**

Portfolio

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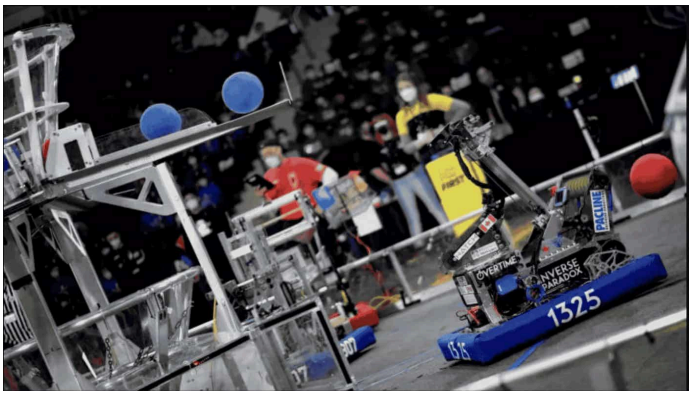
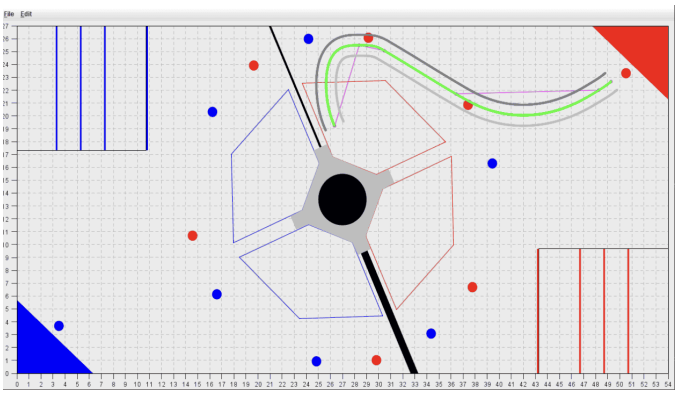
McMaster Rocketry Team, Controls Executive Management

- Led controls sub-team coordinating integration across 6 engineering sub-teams to develop SRAD avionics suite (VLF3 telemetry/DAQ + VL8 power/pyro/camera controller) with STM32 microcontrollers, barometer, and LoRa telemetry.
- Developed visualizer displaying telemetry data with plots(battery voltage, acceleration, altitude), navball orientation display, and view showing rocket location and speed relative to launchpad.
- Co-authored IREC Spaceport America Cup 2023 Project Technical Report documenting avionics architecture, sensor integration, and assembly/preflight/recovery checklists for competition submission.
- Contributed to avionics design supporting future active control extensions through sensor fusion with Kalman filter implementation for powered flight state estimation.
- **Technical documentation available here and video of launch available here**

FIRST Robotics Team 1325 – Inverse Paradox, Software





- Architected autonomous navigation with path following and encoder/gyro feedback enabling shooting while moving, implemented Limelight vision processing with trigonometric calculations and cached position tracking maintaining target lock during sensor dropout.
- Designed state machine for 11-second automated climb with pitch based transitions and soft stops, implemented trigonometric feedforward gravity compensation.
- Developed vision-guided turret with interpolated RPM speed map and OpenCV colour detection enabling automatic cargo rejection, and manual clear.
- **Technical documentation available here and video of a match available here**